

How Much Do High Schools Matter?

Causal Effects on Achievement and Major Choice

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Motivation

Question

- How much do high schools matter for students' later achievement and higher-education choices?
- The school-effects literature has strong evidence on achievement, but much less on whether schools affect field of study and the quality of the higher-education match. (Angrist et al. 2016; Dobbie and Fryer 2015; Angrist et al. 2017; Angrist et al. 2024)
- The practical challenge is that “school attended” is a high-dimensional treatment: one treatment for each high school.

What Changed Since the March Version

- March version: replicate the Chilean assignment system and use lottery offers/risk to estimate school-specific reduced-form effects.
- Spring pivot: replace the many-school treatment with a scalar school-value treatment.
- Current version: make the scalar design computationally tractable with expected-risk controls and reduce noisy school-value estimates with Empirical-Bayes shrinkage.
- The outcome set now extends beyond scores and STEM enrollment to higher-ed enrollment, accreditation, and program-income measures.

Original Problem: Many Treatments

- A fully flexible treatment would include one indicator for each high school:

$$D_i^{\text{full}} = (\mathbf{1}\{S_i = 1\}, \mathbf{1}\{S_i = 2\}, \dots, \mathbf{1}\{S_i = J\}).$$

- The lottery shifts students across schools, but generally does not identify a separate causal effect for every school.
- Students also have different next-best alternatives, so target-school effects are not automatically comparable.

Different Fallbacks

- If all students had the same guaranteed fallback G , lottery-induced comparisons could be interpreted as

$$\beta_{AG}, \quad \beta_{BG}, \quad \beta_{CG}.$$

- In practice, students have different fallback schools:

$$G_i \in \{G^1, G^2, \dots, G^K\}.$$

- The relevant objects become target-school-by-fallback effects:

$$\beta_{AG^1}, \beta_{AG^2}, \dots, \beta_{CG^K}.$$

- This motivates projecting school attendance into a scalar school-value index.

A Related Move: Next-Best Effects

- Kirkeboen et al. (2016) study field-of-study effects using centralized assignment in Norway.
- Their solution is to collapse programs into broad fields and estimate next-best effects across fields.
- The cost is that their main specification abstracts away from institution-level variation.
- My current approach keeps school-level variation, but projects it onto a pre-specified continuous margin.

Next-Best Effects in Practice

Table 5. 2SLS estimates of the payoffs to field of study (USD 1,000)

	Next best alternative (k):								
	Humanities	Soc Science	Teaching	Health	Science	Engineering	Technology	Business	Law
Completed field (j):									
Humanities		21.38* (10.97)	-4.72 (9.85)	-22.93* (12.12)	4.97 (11.86)	-38.51** (14.72)	6.87 (48.29)	-42.21** (10.56)	-156.33 (437.28)
Social Science	18.72** (6.73)		9.84 (11.55)	-10.82 (13.00)	55.46** (21.45)	-55.36** (20.60)	-110.38 (102.97)	-28.37** (10.66)	-76.07 (86.42)
Teaching	22.25** (4.96)	31.37** (7.88)		1.82 (6.55)	23.46** (9.45)	-33.94** (12.54)	-35.32 (37.07)	-21.08** (7.12)	22.78 (127.87)
Health	18.75** (6.25)	30.69** (7.56)	7.72** (2.82)		28.87** (7.64)	-27.87** (10.35)	-43.38** (20.84)	-17.39** (3.97)	-55.19 (97.68)
Science	53.71** (18.37)	69.59** (22.36)	38.58** (14.20)	29.63** (11.53)		-2.21 (14.60)	16.81 (18.07)	-4.92 (10.51)	148.26 (276.20)
Engineering	59.81 (50.59)	-5.53 (58.17)	75.24** (37.50)	0.16 (16.36)	52.35** (20.98)		-46.00 (43.89)	-13.03 (23.70)	-57.66 (166.60)
Technology	41.87** (10.84)	58.69** (10.09)	22.08* (12.44)	32.45** (10.09)	68.07** (9.63)	-5.56 (11.95)		7.03 (9.49)	-53.07 (147.53)
Business	48.13** (11.25)	61.93** (12.03)	31.02** (8.78)	30.22** (10.86)	58.01** (10.48)	-3.42 (12.61)	28.54* (15.61)		3.53 (83.04)
Law	46.34** (7.16)	55.62** (8.34)	36.60** (11.56)	21.49* (11.46)	40.07** (9.68)	-27.53 (18.29)	-15.55 (17.96)	-1.36 (8.66)	
Medicine	83.34** (9.76)	79.39** (10.65)	62.62** (9.02)	45.57** (7.01)	81.31** (9.71)	21.07 (20.67)	40.07** (11.72)	23.34** (8.79)	14.82 (83.61)
Average y^k	30.01	23.40	46.15	51.79	27.31	87.85	78.37	75.61	105.83
Observations	8,391	11,030	10,987	3,269	6,422	3,085	1,245	4,403	1,251

Note: From 2SLS estimation of equations (14)-(15), we obtain a matrix of the payoffs to field j as compared to k for those who prefer j and have k as next-best field. Each cell is a 2SLS estimate (with st. errors in parenthesis) of the payoff to a given pair of preferred field and next-best field. The rows represent completed fields and the columns represent next-best fields. The row labeled average y^k reports the weighted average of the levels of potential earnings for compliers in the given next-best field. The final row reports the number of observations for every next-best field. Stars indicate statistical significance, * 0.10, ** 0.05.

Main Object

- For each school s , construct a school-value index V_s^m for metric m .
- A student's endogenous treatment is the value of the school they attend:

$$A_i^m = V_{S_i}^m.$$

- The instrument is the value of the school offered by the lottery:

$$O_i^m = V_{R_i}^m.$$

- The IV asks whether lottery-induced movement toward higher-value schools causally improves outcomes.

Data and School Values

- Universe: students observed in regular grade 8 in Chile between 2017 and 2020.
- Approx. 944,000 unique students.
- Linked records:
 - grade-4 SIMCE math and language scores;
 - grade-8 cohort, gender, age, and home comuna;
 - centralized assignment probabilities and lottery offers;
 - high-school enrollment histories after grade 8;
 - PSU/PAES math and language scores;
 - higher-education enrollment, fields, accreditation, and program-income measures.

Assigning a High School

- For each student, I track school enrollment during the four years after grade 8.
- Main school assignment:

S_i = school code attended for the most post-grade-8 years.

- Ties are broken toward the most recent school.
- This maps each student in the broad grade-8 universe to one high-school code when available.

Raw and Adjusted School Values

Raw school mean

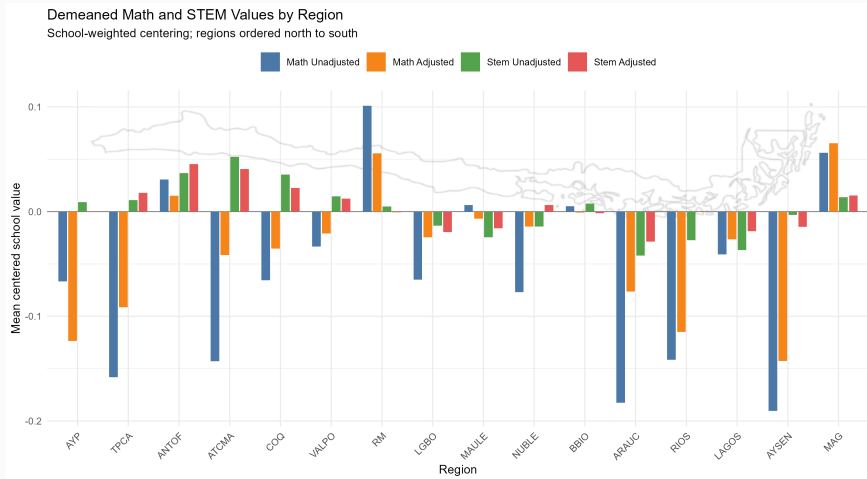
$$V_s^{\text{raw},m} = \frac{1}{N_s} \sum_{i:S_i=s} Y_i^m.$$

Adjusted observational value-added

$$Y_i^m = \mu + V_{S_i}^m + \sum_{p=1}^3 \lambda_{Mp} M_i^p + \sum_{p=1}^3 \lambda_{Lp} L_i^p + \gamma_{\text{cohort}(i)} + \delta_{\text{gender}(i)} + \rho_{\text{age}(i)} + \kappa_{\text{comuna}(i)} + \varepsilon_i.$$

- V_s^m is a school fixed effect, interpreted as an observational and potentially selection-contaminated school-value metric. (Chetty et al. 2014; Angrist et al. 2017)

Adjustment Matters



Current IV Design

Lottery Sample

- I use students who enter the grade-9 centralized assignment mechanism (*SAE*) on time.
- I restrict to students with non-trivial assignment risk:

$$\max_{s \in \mathcal{S}} p_{is} < 1,$$

where p_{is} is the simulated probability that student i is assigned to school s .

- Current diagnostics: about 419,000 student-process rows, 163,600 timely at-risk rows, and 106,374 exam-taker timely at-risk rows.
- The offer provides exogenous variation in attended-school value, conditional on lottery risk.

Previous Risk Controls

- The spring design used a supported school set and included two controls for each supported school:
 - the probability of assignment to the school;
 - an indicator for zero assignment probability.
- With the 100-student support threshold, this meant roughly 2,000 risk controls.
- This was feasible for main estimates but costly for heterogeneity and extensions.

Current Risk Control: Expected School Value

$$E_i^m = \sum_s p_{is} V_s^m.$$

- p_{is} : simulated probability that student i is assigned to school s .
- V_s^m : school value for metric m .
- Instead of controlling for the full probability vector, I control for the scalar expected value of the lottery risk set.
- This keeps the design close to the assignment-risk logic while making the current regressions much easier to run and extend.

$$A_i^m = \pi_0 + \pi_1 O_i^m + \eta E_i^m + X_i' \lambda_D + u_i.$$

$$Y_i^m = \alpha + \theta \hat{A}_i^m + \phi E_i^m + X_i' \lambda_Y + \varepsilon_i.$$

- A_i^m : value of the school attended after grade 8.
- O_i^m : value of the first-round offered school.
- E_i^m : expected school value under assignment probabilities.
- X_i : cohort, gender, grade-8 age, grade-4 math, and grade-4 language controls.

Main Results

How to Read θ

- The coefficient θ is a pass-through parameter from observational school value to lottery-induced causal gains.
- If a complier is induced to attend a school with one unit higher V_s^m , their outcome is expected to move by θ units.
- If $\theta = 1$, the observational school-value index maps one-for-one into the causal effect along the lottery margin. (Angrist et al. 2017)

Expected-VA Main Results

Specification	Math VA		Language VA		STEM VA	
	θ	SE	θ	SE	θ	SE
Unadjusted	0.246	0.029	0.190	0.029	0.390	0.050
Adjusted	0.841	0.071	0.835	0.082	0.689	0.067

Expected-risk-control specification. Adjusted school values are much closer to one-for-one pass-through.

Interpretation

- Observational school value predicts lottery-induced causal gains in achievement and STEM enrollment.
- Adjusted metrics perform substantially better than raw school means, consistent with selection bias in unadjusted values.
- The STEM result suggests that high schools affect higher-education major choice, not only test scores.
- The expected-risk-control design makes it easier to run heterogeneity and new-outcome exercises.

Heterogeneity by Grade-4 Math Quintile

Grade-4 math quintile	Math VA		Language VA		STEM VA	
	θ	SE	θ	SE	θ	SE
Q1	0.145	0.145	0.596	0.180	0.735	0.132
Q2	0.785	0.133	0.601	0.178	0.487	0.144
Q3	0.940	0.151	0.929	0.188	0.955	0.153
Q4	0.771	0.158	1.122	0.182	0.567	0.166
Q5	1.317	0.166	0.891	0.184	0.707	0.163

Adjusted expected-VA estimates, full grade-4 SIMCE split.

Higher-Education Quality Outcomes

Specification	Program-certified years		Institution-certified years	
	θ	SE	θ	SE
Unadjusted	0.299	0.115	0.688	0.112
Adjusted	0.872	0.156	1.153	0.123

The scalar school-value framework also predicts lottery-induced movement into more accredited higher-education options.

Empirical-Bayes School Values

Why Shrink School Values?

- School value-added estimates are noisy, especially for smaller schools and rarer outcomes.
- The current EB path shrinks first-stage observational school values before using them as the scalar school-quality index.
- This is not a new causal estimand. It validates an EB-shrunk observational school-value index along the SAE lottery margin.
- EB and non-EB θ 's should be interpreted together with the scale of the underlying school-value index.

$$\tau_m^2 = \max \left\{ \widehat{\text{Var}}_w(\widehat{V}_s^m) - \widehat{\mathbb{E}}_w \left[\widehat{\text{se}}_s^2 \right], 0 \right\}.$$

$$\lambda_s^m = \frac{\tau_m^2}{\tau_m^2 + \widehat{\text{se}}_s^2}.$$

$$V_{s,EB}^m = \bar{V}^m + \lambda_s^m \left(\widehat{V}_s^m - \bar{V}^m \right).$$

- Weights use the number of students in the school-value regression.
- Output is recentered to preserve the student-weighted zero-mean convention.

$$A_i^{m,EB} = V_{S_i,EB}^m, \quad O_i^{m,EB} = V_{R_i,EB}^m, \quad E_i^{m,EB} = \sum_s p_{is} V_{s,EB}^m.$$

- The IV specification is the same expected-risk-control design.
- The only change is replacing the raw adjusted school-value index with the EB-shrunken index.
- The mean assignment-probability mass covered by EB values is about 0.865 across the main metrics.

EB Main Results

	Math	Language	STEM enrollment	Institutional quality
θ^{EB}	0.881	0.908	0.824	1.508
SE	(0.074)	(0.089)	(0.080)	(0.166)
Observations	90,906	92,472	92,959	92,959

EB-shrunken values preserve the core finding and make the main pass-through estimates close to one for test scores and STEM.

Additional EB Outcomes

	Exam taking	Higher-ed enrollment	Program income
θ^{EB}	0.478	0.748	1.316
SE	(0.053)	(0.117)	(0.154)
Observations	135,898	135,898	92,959

Current evidence points to effects on participation in the higher-education pipeline and on the value of the eventual program match.

Program-Income Components

	Program income: area FE	Program income: institution FE	Program income: full
θ^{EB}	1.118	1.327	1.084
SE	(0.144)	(0.167)	(0.133)
Observations	92,959	92,959	92,959

The program-income exercise separates broad field composition from institution-level and full program-income measures.

Decomposing Full Program-Income VA

- I decompose school-level full program-income VA into orthogonal components.

- Current preferred ordering under consideration:

exam taking $\rightarrow$ math $\rightarrow$ area income $\rightarrow$ institution income.

- This asks whether schools shift students toward higher-income programs through exam participation, achievement, field composition, institution choice, or residual channels.

Decomposition: Exam-First Version

	Exam-taking component	Math component	Institution component	Area component
θ^{EB}	-2.003	0.426	-0.022	1.152
SE	(0.566)	(0.435)	(0.762)	(0.130)
Observations	92,958	92,958	92,958	92,958

Preliminary scalar-component results suggest the area/field component is the clearest positive channel in this ordering.

Where This Leaves the Project

Current Takeaways

- The scalar school-value approach makes the assignment lottery usable without estimating one treatment effect for every high school.
- Expected-VA risk controls make the design substantially more scalable.
- Adjusted and EB-shrunk school values predict lottery-induced gains in scores, STEM enrollment, accreditation, and program-income measures.
- The results are consistent with high schools affecting higher-education trajectories beyond achievement alone.

Remaining Interpretation Work

- Clarify exactly what pass-through parameter is identified when the school-value index is noisy or shrunken.
- Compare EB and non-EB results in units that account for the SD of each school-value index.
- Decide whether the expected-VA risk-control design is the main specification or a scalable complement to the high-dimensional risk controls.
- Refine the program-income decomposition and decide which component ordering is most defensible.

Next Short-Term Steps

- Formalize the expected-risk-control estimand and its relation to the earlier probability-control specification.
- Add scale diagnostics for EB versus non-EB school-value indices.
- Finish pooled interaction tests for gender and baseline-achievement heterogeneity.
- Turn the program-income decomposition into a more explicit mechanism exercise.

Thanks!